Benchmark Comparison: AFFORD-Bench vs. LIBERO-PRO & LIBERO-X

Key Differences in Physical Affordances, Perceptual Modalities, Policy Universality, and Scale

RSS 2027 BENCHMARK

BENCHMARK DIMENSION Core Evaluation Axes	LIBERO-PRO Mellis et al., 2024 • Prior Baseline	LIBERO-X Kim et al., 2025 • Language-X	★ AFFORD-Bench (Ours) Physics-Grounded Universal Benchmark
Contact Physics Friction & mass dynamics	Nominal (Fixed μ = 0.5)	Nominal (Fixed μ = 0.5)	✓ Dynamic μ [0.1–1.8] & Mass
Perception Modalities Visual & geometric inputs	1× RGB (128 × 128)	2× RGB (256 × 256)	✓ Dual RGB (256²) + 3D Depth
Policy Universality Policy architecture support	VLA Models Only	VLA Models Only	✓ Universal: DP, ACT, Flow, VLA
Domain Scope Environment breadth	4 Domestic Suites	5 Domestic Suites	✓ 7 Diverse Domains
Task Complexity Horizon & subgoals	Single-Step Pick & Place	Subtask Tagging	✓ Chained 2–3 Step Subgoals
Demonstration Scale Total steps & trajectories	1.3M Steps (6.5k Demos)	0.5M Steps (2.5k Demos)	✓ 2.3M Steps (10.5k Demos)
Benchmark Scale Normalized data volume	130 Tasks • 56% Scale	50 Tasks • 22% Scale	★ 150 Tasks • 100% (2.3× Scale)